SECURESHIFT

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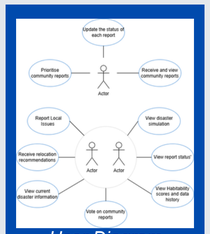
SecureShift is a web-based platform that evaluates neighbourhood habitability and enables community issue reporting, helping users understand safety, sustainability, and access to essential services through a clear habitability score. It supports SDG 11, particularly Targets 11.1, 11.3, 11.5, 11.6, 11.7 and 11.b, by improving access to safety information, enabling community participation, increasing disaster awareness, highlighting environmental issues, and supporting more resilient and inclusive urban development.

Milestone 1 Reflection: Project Proposal

The original aim of the prototype was to expand the capabilities of existing platforms by combining multiple sources of neighbourhood information into a single system. SecureShift uniquely integrates area-specific data with community-driven input to provide actionable insights. This approach improves how users understand safety; sustainability; and local conditions as well as supporting more informed decision-making. Through SecureShift, the users are able to contribute to a safer; more inclusive; and more sustainable way of urban living. It successfully demonstrates the core solution by simplifying complex neighbourhood data into clear habitability scores and enabling community reporting.

Milestone 2 Reflection: Software Requirements

Prototype 1 of SecureShift implements the key features required, such as a reporting tool (6); a habitability scoring dashboard (3); an authority dashboard (8) and a living standard forecast (1), while prioritising accessibility (5) needs for all users. Throughout the development, there have been some changes made to the implementations of these features, such as the authority dashboard being able to promote and demote user access levels, only current (prepopulated) data scores to be displayed; the standard of living forecast to give general guidance based on user input and for reports to include threads of comments made by multiple users. The features not implemented were so that we could focus on the core functionalities of this system for the first prototype.Through these implemented features we successfully address the User Stories defined in this milestone, including User Stories 1; 3; 4; 5; 6; 7; 8 and 13. The further functionalities of these features, and the excluded features, are to be implemented in prototype 2.



User Diagram

Milestone 3 Reflection: Design Plan

The feature selected to fully and partially implement in this milestone have successfully implemented the core functionality of the system and align with the user needs. Of the features to be fully implemented, Accessibility has been correctly implemented in its entirety. For the Habitability overview, there was a change implemented to not include the interactive map here, but rather in the Reporting feature. Of the partial implementations, the Reporting feature does not include the prioritisation/ filter design in this feature and instead includes threads of user comments. The main implementations of the authority dashboard have been moved to a prototype 2 design, although they can now promote and demote users. The Habitability Scoring and Forecast features now use prepopulated datasets for this prototype to reduce complexity, and still successfully work as intended. There is now a login feature for every user, where they can login or register. If registering for the first time, then a One Time Password will be sent out to them.The chosen technologies (Vue, Django, REST API), excluding PostgreSQL, have been used effectively and supported efficient development and system integration. We decided to use SQLite for the database due to its seamless integration into Django and because it is more lightweight than PostgreSQL, which is better suited to disaster-prone areas.

Scoring Calculations

Architecture Model

Milestone 4 Reflection: Software Development

As a team, we coordinated the development by dividing responsibilities across frontend, backend, database, and DevOps roles based on individual strengths:

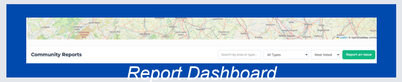
- Phoebe McDonald - database models for Habitability and reports
- Charlie Homer - Frontend and backend of habitability score comparison, and backend of user registration, OTP email verification, and login
- Haider Shah - Frontend pages for Authority and Forecast views.
- Muhammad Zayan Zahid - Planning with refinements for database models for reports and users
- Chloe Cheah - frontend for login, registration, home and reports view.
- Iman Asyik - Accessibility Features
- Yahya Daya - Backend for reports and comments models, endpoints for comments, user storing/changing details logic.

Throughout the development process, we continuously met up as a group to discuss the progress we had made and the next steps that were needing to be taken. Within the sub division groups for development, we kept each other updated with what work had been completed and kept communication flowing. Within the team, all commits and merges on GitLab were successful. Minor issues included resolving integration issues, particularly when ensuring consistent data flow between components.

Testing & Results

Testing included both unit and integration testing to ensure core functionality operated correctly. Key workflows such as viewing habitability scores, displaying map data, submitting reports, and updating report status were successfully tested and functional. Results showed the system operates as intended for prototype 1. These results highlight the importance of early testing and iterative development, informing improvements for future prototypes, particularly in feature completeness and system robustness.

Forecast Dashboard



Report Dashboard

Habitability Score Dashboard

Prototype 2 Implementations

The prototype 2 implementation of SecureShift will build upon the core features that were implemented and designed during this cycle. This includes building upon features such as the Authority dashboard to include access to update the status of a report. Alongside this, the Habitability Score dashboard will be expanded to include visual representation of historical data for users to view and live data for the calculations to be updated with. Prototype 2 will also include more advanced features such as the Forecasting feature to predict (based of historical data) how the user's area will vary over time and relocation recommendation services as mentioned early on in the project proposal.